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Personal networks typologies: A structural approach

Claire Bidart[a,*], Alain Degenne[b], Michel Grossetti[c]

[a] Aix Marseille Univ, CNRS, LEST, UMR 7317, 35 Avenue Jules Ferry, 13626 Aix en Provence, France
[b] CNRS, 2 rue des Fusillés, 37000 Tours, France
[c] CNRS, EHESS, Université de Toulouse UT2J, LISST-Cers, UMR 5193, 5 allées A. Machado, 31058 Toulouse Cedex 09, France

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ABSTRACT

Building typologies allows to compare networks on multiple dimensions, and to approach a generalization grounded on empirical data. In this article, we present a typology of personal networks only based on indicators related to the structure of relations between alters. It is designed from very detailed data on young French people who were involved in a longitudinal study. Our typology mobilizes a small number of indicators to discriminate the types that compose it. In so doing, we intend to make it applicable to various surveys.

1. Introduction

How can personal networks be compared? Social sciences are based upon comparison: between contexts, between categories of individuals, between periods, and so on. Network analysis has shown its great potential for the study of intermediate social forms and their dynamics. It has developed many indicators providing very relevant information for social scientists: the size of the network, its composition, density, centrality, modularity, and so on. These indicators help describe networks systematically.

The information they provide has very specific sociological meanings that contribute to the understanding of the multiple ways in which a person's social circles develop and get structured. Whether the network is extended or concentrated, homogeneous or heterogeneous, dense or segmented enlightens us about the forms of socialization that constitute, according to Simmel, the very object of sociology (Simmel, 1950). In this perspective Burt (1992), for example, in his analysis of personal networks, highlights the fact that the central person of this network, ego, manages its diversity and controls potential exchanges between alters who make up the network. These degrees and arrangements of connections between alters in a network form its structure. In the field of analysis of the impact of network structure, various indicators are thus used to describe this structure.

In this article, we focus on personal networks, for which connections between alters are described by ego, while for complete networks each network member is interviewed and asked to describe his own connections. As McCarty (2002) reminds us, networks of ties between alters of the same ego can be analyzed using the same techniques as for the "complete" networks. Some indicators relate to the scope of the personal network and its composition. The aim is to evaluate the "social surface" of a person by measuring (and comparing to others) the number of alters in the network and their diversity. The latter indicates the multiplicity of social circles to which that person has access. The issue of homophily has become a classic question of network analysis in social sciences (McPherson et al., 2001; Di Maggio and Garip, 2012).

Other indicators focus on the overall personal network structure. This is the case for density, which indicates its general level of cohesion, depending on the extent to which alters are interconnected.

More complex indicators relate to different parts of the network. One approach is to highlight dense areas in the network; these are referred to with the ambiguous but now widespread term of "communities". Various indicators have been constructed in order to assess to what extent such communities differ from their neighborhoods in the network (Blondel et al. 2008; Newman and Girvan, 2003). Finally, structural indicators may be applied to individual alters. This is the case for various kinds of centrality (degree centrality, betweenness centrality, proximity centrality), which show that some alters occupy a position that gives them a specific role. One can measure the distribution of these centralities within the network, to obtain a comprehensive indicator of its global structure.

Having stable typologies of personal networks based on structural criteria becomes more and more interesting given the increasing amount of available data generated by online activities, in which little information beyond the networks and their struc-

* Corresponding author.
E-mail addresses: claire.bidart@univ-amu.fr (C. Bidart), anicalain.degenne@numericable.fr (A. Degenne), Michel.Grossetti@univ-tlse2.fr (M. Grossetti).

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tures is given. It also facilitates comparisons between the different surveys on personal networks.

In this article our aim is to develop a basis for a systematic comparison of personal networks articulating different perspectives while giving a synthetic view of the structure of these networks. We undertook the task of building structural typologies by seeking combinations of criteria which allow us to efficiently sort personal networks and to assign them to the most relevant types. To do so, we rely on a longitudinal survey of personal networks of young French people living in the Caen area, as they enter adulthood. We have already explored the biographical dynamics of these networks from various perspectives in previous studies (Bidart and Lavenu, 2005; Degenne and Lebeaux, 2005; Bidart and Cacciuttolo, 2013; Bidart, Degenne and Grossetti, 2011).

We are now using this empirical investigation to elaborate a typology of personal network structures. This typology should be sufficiently complex and account for enough characteristics for the social scientist not to stray too far from the social reality, which is itself complex; but, ideally, it should also be simple enough to suggest applications to larger numbers of personal networks stemming from various surveys. Our knowledge of these data enables us to construct a typology that is realistic. Naturally, this is a first step and this typology will be tested on other datasets.

We first mention some contributions to the study of network structures and clarify the relevance of some indicators commonly mobilized in descriptions and typologies. We then depict the data collected by the longitudinal qualitative study we are mobilizing, which was conducted in France in four waves of survey over the course of nine years. We then present our own progression and technique in order to bring out the fundamental characteristic traits we retain for constructing an inductive typology. We unfold the steps of the analytical work which allows us to characterize the personal networks in this survey in a simple manner and compare them. To give it a heuristic value, we then seek to find out what indicators and thresholds enable us to account as accurately as possible for their structures. Our aim is to propose a reproducible way to synthesize personal network structures via these types.

We check the relevance of this typology by measuring its distribution based on a key variable for the social sciences, namely social class, which is measured here by the parents' occupation and the educational pathways of respondents.

Based on empirical data, this typology is therefore meant to be an attempt to produce structural measures of personal networks that are realistic and can be applied generally.

2. Typifying personal networks

The large numbers of surveys on personal networks have stabilized recurring results such as the correlation between network size and level of education, or the importance of family ties. There are fewer stable results on the structural characteristics of personal networks, even though we know that such characteristics are closely linked to social situations and to the composition of networks.

When focusing on the network structure, one must seek to gather information on the links between the alters cited by ego (the respondent). The most common method is to show the respondent the list of names obtained through the name generators, presented as a matrix, and ask him/her who is related to whom. The result is a network of nodes and links between them, to which the same measures can be applied as to a complete network. McCarty (2002) used the same methods for analyzing personal networks as for complete networks because they allow the analyst to take into account the structure of the relational system. However, when the number of alters is low (in classic surveys by Wellman (1979, 1981) and Marsden (1987) for example, this number is about four on average), one can hardly engage in detailed structural analyses. For that reason we are using a survey which does not limit personal networks' size and thus allows a wider scope and diversity of structures.

The place given to structural measures in network analysis is variable. Some typologies are based on the composition of the network in terms of alters' characteristics (age, level of education, occupation, and so on) or in terms of characteristics of ties (duration, strength, multiplexity, and so on); others involve some network structure indicators; yet others adopt a purely structural perspective, and only afterwards analyze the links between the resulting typology and other variables. We present here some of these attempts to build personal network typologies.

Typologies based on the attributes of alters are the most common. In a recent meta-analysis of personal networks, researchers identified 277 studies whose results they were able to reexamine in order to analyze changes in networks and life events (Wrzus et al., 2013). To do so, they mainly used the composition of networks rather than structural parameters, because the field of such an analysis was wider.

Thus, for example, Clare Wenger (1991) studied seniors' networks, in terms of the support and care they can provide. Personal networks were classified according to the proportion of family members, friends, neighbors or others such as members of associations. In such a study, alters were mainly identified in terms of role.

Agneessens et al. (2006) also studied networks of older people, but they focused on the support functions of alters (emotional, instrumental and accompanying). They used a method of latent class to sort the networks according to the distribution of alters in these types. They retained a model made up of three types: "no support", "companionship" and "companionship and emotional support." Among the most recent typologies, Eric Gianella and Claude Fischer (2016) used 21 variables that combine the attributes of the alters woth some structural aspects (size and density).

These studies are just presented here briefly as some examples of elaborations of typologies, in these cases based on the characteristics of alters or their relational roles. Scholars mobilized factors they hypothesized as relevant for their social question, that is the link between social integration of egos and usefulness of their network. All these analyses were using classifications, but network structure was not involved.

In another group of studies, structural indicators were mixed with other indicators.

In some cases, scholars studied the links between structure of networks and the qualities of ties. For example, Haythornthwaite (2000) studied personal networks of 52 students participating in a graduate degree program completed at a distance via the Internet. She asked students to describe their links with the other students in the same class, using categories that refer to the strength of ties (close friend, friend, work only). She did not build a typology of networks, but she examined the correlations between the structural characteristics of the network obtained (composition, density, size) and uses of different types of communication media.

In other cases, structural indicators were mixed with indicators of composition of personal networks.

For example, McCarty (2002) used indicators of structure and position of the networks of 46 individuals, to which he subsequently added a characterization of relational roles. In his survey, each person was asked to list the names of 60 people he or she knows and say whether there was a link between those named. The author thus had 46 adjacency matrices. He then calculated six structural variables for each network: density, measures of dispersion of three kinds of centrality (degree, proximity and betweenness) (Freeman, 1978), an indicator based on the number of cliques, and the number of related components of the network.Based on these elements, he classified the nodes into 'clusters' for each network. Secondly, he asked respondents to characterize these clusters. He thus obtained 12 types of clusters: family, network via other person, couples, network from childhood or people with whom one grew up, school network, colleagues, housemates, network through religious affiliation, hobby groups, issue-oriented groups, neighbors, other sociability groups. In such an analysis, structural indicators were predominant, and the focus of ties served as complementary information for the identification and interpretation of clusters. These two kinds of data were used sequentially.

Lubbers et al. (2007) also studied migrants networks. They classified the networks into five profiles using simultaneously both data concerning the composition of networks and structural data. Each network was described by nine variables: percentage of Spanish people, percentage of migrants, number of cohesive subgroups, density, betweenness centrality, average frequency of contacts, average proximity, and percentage of family members. They obtained five types: small, sparse networks, dense kinship networks, networks with multiple sub-groups, networks connecting two social circles, and rooted networks. Their typology integrated multi-dimensional informations (structural, role-relationship and characteristics of alters in terms of origin). Their combination was integrative.

In an attempt to draw a typological mapping of migrants' networks, Brandes et al. (2010) described the networks of people from the Caribbean, Africa or Eastern Europe who moved to Spain or the United States. The data included information about the migration situations of ego and the alters (fellows, hosts, origins, transnationals), informations about the strength of ties between the alters, and the density of networks. The analysis method was inspired by models-block: each network produced a 4×4 matrix whose rows and columns represented these migration classes and whose cells contained a weighted average of the strength of ties observed between the alters. Each network was thus projected homomorphically on a four summits picture network. These picture networks were then classified by a statistical method of types construction. In a first step, eight types were obtained. They were then grouped into four main types of networks combining these indicators (integration, assimilation, separation, marginalization).

In her thesis Cruz Gomez (2013) focused on the study of the determinants of homophily, especially the role of social status. So she introduced questioning on the characteristics of network members (occupational categories, level of education, origin, gender, age, etc.). But she also made a typology of the network structures before measuring the distribution of the types according to these characteristics. Four variables describing networks were mobilized: density, average degree of centrality, average variation of proximity centrality, and average variation of betweenness centrality. By going back and forth between visual classification and statistical analysis, she identified nine types and named them according to their general shape: cell, daisy, scorpion, lobster, butterfly, patches, comb, ray fish (p.91). She then combined these types and the characteristics of alters composing networks in her analysis. In that case, the elaboration of types is the result of an inductive and iterative process between statistical computation and visual analysis of the networks' shapes.

Such "mixed" typologies combine structural data, either visually described or automatically computed, with data concerning alters, links or their contexts. These combinations sometimes give priority to structural data, sometimes to individual or contextual data, but it is clearly the combination of these dimensions that enables the typology to be completed.

Very few research studies set out to make a strictly structural typology, namely refraining out of hand from taking into account data characterizing the persons that make up the network or their social contexts.

The article by Faust and Skvoretz (2002) may be considered pioneering in that respect. Their purpose was clearly a methodological one. Their empirical basis consisted of 42 networks from four categories of living species – humans, nonhuman primates, other non-primate mammals, and birds. The authors wrote: "The methodology we propose is generally applicable to the characterization and comparison of network-level social structures across multiple settings, such as different organizations, communities, or social groups and to the examination of sources of variability in network structure." (p. 267). In addition, some networks covered the same populations but referred to different relationships; for example, between the monks of the monastery analyzed by Sampson, they distinguished positive influence, negative influence, blame, praise. In a population of chimpanzees, they distinguished three relationships: pant-grunt calls, agonistic relation and grooming.

They used the p^* model to create the conditions for comparability of networks together. Let us recall that the p^* model takes as references a number of elementary forms. Here, they were six which list different possible forms of dyads and triads. The model stated that the probability of a given graph is a log linear function of the number of each of these elementary figures occurring in the graph. It estimated a vector θ which is that of the parameters of this function. On the basis of this vector Faust and Skvoretz founded their comparison. They summarized their results as follows: "It appears that the kind of relation involved rather than the species underlies similarities among the networks. It is the nature of relations that determines the structural features of its network. For example agonistic relations, whether between red deer or highlands ponies are similarly structured. This leads to the speculation that distinctions among species in network structures are due to differences in relations in which they typically engage. (p.290)". Because of the heterogeneity of their populations, the result did not allow to go further towards a typology, but it confirmed the relevance of the question and the method. It clearly focused on the "pure" structure of networks, leaving apart considerations of populations and contexts, even if they considered some characteristics of ties.

As another example, Kennedy et al. (2015) presented a method for constructing and testing cohesion in couples' networks. They interviewed both partners in 57 couples separately (2 interviews in nine months), and they matched their common and separate alters in the "duocentric network." They first visualized the networks, and then examined corresponding quantitative measures of characteristics of the networks. They considered density, the number of components, the number of isolates and the degree centrality. With these simple measures, they showed great contrasts in cohesion, but they didn't make a typology. They integrated a visual stage in their analysis, combined with measures of structural indicators.

Backstrom and Kleinberg (2014) also depart from a purely structural method for identifying the strongest links, particularly spouses, in networks of Facebook users. Their database consists of 1.3 million Facebook users, and a smaller dataset of 73,000 neighborhoods from this first collection. Thus the scale changes.They rely on a recursive measure of the dispersion of networks for identifying, within them, love partners (or best friends or close relatives). Their measure combines network closure and brokerage between groups, observing the number of mutual Facebook "friends" but also the structure of the neighborhoods of these mutual friends: "Roughly, a link between two people has high dispersion when their mutual friends are not well connected to one another"(p. 832). This identifies the "signature" of strong ties sharing several contexts with ego better than the measure of embeddedness does. They then compare the performances of different measures with the true partners (romantic, engaged and married people). In this analysis also, priority is given to structural measures, even if the later stage allows to check their relevance.

This brief overview allows to underline some characteristics of network analysis in terms of typology, and to specify our own approach. Rather than focusing on composition of personal networks, or mixing structural data with alter or ties characteristics, we propose a "pure" structural analysis with a classification, and then a final checking of the social relevance of classes. We also include a visual and descriptive stage and a back and forth process in building the structural types, followed by an attempt to "translate" these types into precise thresholds for a limited number of indicators. This inductive then analytic approach aims at constructing some reproducible tools for further generalizations.

3. Empirical elements for a structural typology

3.1. The data

A longitudinal qualitative survey of a panel of young people was based on a series of interviews conducted with the same young people in four waves at intervals of three years.[1] In 1995, 87 young people living in the Caen urban area (Normandy, France) were interviewed. One third were in the final year of high school studying for the economic and social baccalauréat, one third in the final year of preparation for the vocational baccalauréat and one third were taking part in a labor market integration program. Each group contained equal numbers of young men and women. They were aged between 17 and 24. In the second wave in 1998, 75 young people responded again, 66 of them responded in 2001, and 60 in 2004. We thus obtained a total of 287 personal networks.

In each wave of the survey, the process of constructing each of the personal networks was carried out by means of an original context-focused name generator (Bidart and Charbonneau, 2011). A long series of questions on life's different contexts has to cover as broadly as possible the contexts young people may dwell in: school, university, work, recreational activities, various organizations (e.g., associations, political movements, religious organizations), informal groups of friends, the neighborhood, people met through a spouse, people met through the Internet, and so on. The contexts include past as well as present contexts—ones in which the respondents no longer move, but in which they knew people with whom they are still in contact: former school friends, colleagues from previous jobs, old friends, former neighbors, childhood friends, army friends, people met through former love relationships, and so on. Deceased persons and animals were not listed. In all, 53 contexts in which individuals can potentially move (or have moved) were systematically cited in this way. It served to compile the lists of individuals (alters) the respondents (ego) kept company with at different periods of his/her life.

For each context, respondents were asked: "In (this context), have you met people whom you know a little better, with whom you speak a little more?". A list of first names of alters was then compiled. Two screening questions per context were added; they allowed the interviewer to differentiate strong links from weak ties: (1) "Do you see any of them outside of (the context-e.g., work)"? (2) "Are any of them important to you, whether you see them elsewhere or not?". The individuals mentioned in response to one of these last two questions were regarded as strong ties, the others as weak ties. We then asked respondents to say for each alter if he is related or not with others. The average size of the resulting networks is 38 alters, but the differences are considerable: the smallest network has 6 alters, the largest has 134. These variations are especially sensitive to the social class of origin: the average size is 46 alters in the upper classes, in the middle classes it is 42 alters, and 33 alters among the lower classes.

Taking as the corpus the 287 networks of our survey, including all survey waves, we are seeking to describe the structures of these networks. We shall use no characterization of nodes or edges that is not strictly structural.

3.2. Search for indicators

One can compute a large number of structural indicators on a network. Some of them have a fairly intuitive meaning, others are more technical, but in any case those of concern have an implication for social sciences. So we must find a way to identify these structural features technically. In this presentation, we consider the networks of reciprocal ties between alters of the ego (ego not included in the network). We are deliberately limiting ourselves to the indicators we will subsequently mobilize. Given our extensive knowledge of the data, we decided to start with a qualitative and visual network analysis to develop an initial typology. We then realized that four structural characteristics of the networks in our survey (density, centralization, segmentation, dispersion) clearly characterized a good proportion of networks and we tried the experiment to construct a typology based on this. This led us to introduce another variable, diameter, which adds information on the chain length between the dense parts, that is an interesting aspect of personal networks structure.

3.2.1. Density

It is defined as the ratio between the number of existing ties divided by the total number of possible ties, or $n(n-1)/2$ (in the networks we are studying, ties are not oriented, as relationships are assumed to be reciprocal). Density can be measured in the case of all networks, but the more complex they are, the less this information is relevant because it "smoothes out" internal differences. Nevertheless, density is a key indicator because it is remarkably synthetic. In personal networks, the density tends to decrease when the size of the network increases, which is a problem for a typology like the one we seek.

3.2.2. Betweenness centralization

Betweenness centrality applies to nodes of the graph; it assesses how essential it is to go through one node if any two other nodes are going to be able to connect to each other by following the edges.

It is a measure of a kind of power linked to the position of one of the alters, who is a good mediator between others. But this measure is calculated for each node. However, what concerns us here is to see whether, at the level of the alter–alter network, a node or a group of nodes is particularly central and plays almost the same role as ego. To answer this question, we shall use an indicator of the concentration of the distribution of betweenness centralities; it therefore characterizes the alter–alter network. We call it betweenness centralization.

3.2.3. The segmentation of network into social circles

Is the network divided into separate components? This question is important for social scientists, as it infers that a relatively disparate network can remain so, since these components do not communicate with each other. Ego can then let people with different viewpoints co-exist in his network, without risking confrontation between them. He will also get non-redundant information and will be able to distribute his commitments himself, without letting them contaminate each other. This question may refer to several structural aspects of networks:

- The existence of several components that are internally related but separated from each other. Some people arrange their family

[1] For further information see: http://panelcaen.hypotheses.org/.

on one side, on the other side their colleagues, on a third side their friends, and no tie is observed between these different sets of people. These are totally disjointed circles.
- A variation of this model is that ego has a main connected component on the one hand, and on the other hand some small components of isolated individuals or couples who are not connected to this central core.
- A third case, which is not as easy to identify, is that of a connected network in which some denser areas appear, which are called communities. For example, there will be on the one side ego's family in which everyone knows almost everyone; in another part of the network will be colleagues, some of whom know some family members; in other parts there will be neighbors or friends, some of them connected with family members or colleagues, and so on.
- Finally you can, and often do, encounter mixed structures that combine the above cases.

The identification of connected components is easy to operationalize. It is the same with isolated alters. One may also wonder whether it could be relevant to consider isolated alters as specific groups, an approach that pushes the logic of networks as social circles to the limit: each individual is a piece of society.

Detection of communities requires using algorithms that make the procedure less intuitive and less controllable. There are two kinds of algorithms, algorithms through segmentation which start from the total network and cut it up, and algorithms through aggregation which start from nodes and bring them together in order to maximize a certain criterion.

A simple algorithm through segmentation is presented by Newman and Girvan (2003). They start with calculating a betweenness index for each edge of the graph. It is based on the proportion of the shortest paths between two nodes which go through this edge. Edges are weighted by this index. The edges that are strongest in betweenness centrality are therefore candidates for deletion in order to segment the network while preserving communities. There are many algorithms for switching from such a matrix to a segmentation tree. This method has the virtue of being intuitive and provides good results, but it is very time consuming for computation and thus it is not adopted in software dealing with large networks.

We selected the method programmed in Pajek and Gephi, which is known as the method of Louvain (Blondel et al., 2008). The algorithm proceeds by aggregation, maximizing the modularity index. It can be computed from a partition of the nodes of a graph. It is defined as the difference between the share of edges that connect nodes within communities, and the expected value of that quantity in the case of a random graph of the same size and partitioned the same way. This index provides a relevant indicator, the network being connected or not. This is one of its advantages. Another advantage is that it is not dependent on the network size.

3.2.4. Diameter

In a relational network, it is interesting to assess the degree to which two individuals can communicate. The path they have to cross in the graph gives an indication. The diameter is the maximum length of such a geodesic path. We retain it because it allows us to characterize the "stretched" networks, a profile that could be related to the existence of structural holes. We then tested different arrangements of these indicators for building a typology.

4. Towards a typology

4.1. A visual stage

The idea was to start from visual recognition (the network looks like this, differs from that) to establish an initial classification that was entirely empirical. We used the Kamada-Kawai algorithm visualization of Pajek. In doing so, we trusted both our eyes and our knowledge of "sociologically relevant" indicators (Freeman, 2000). For example, we considered that having two important components in a network was quite different from having only one, even one that is surrounded by a few isolated individuals, because the network is then split into two unrelated parts. Similarly, at equal density, some networks are very focused on a particular central alter, while in other networks connections are more evenly distributed. Given that a single indicator cannot summarize these significant differences, we preferred to start with the same descriptive features and only later try to systematize the measurement.

This first inductive classification led us to adopt six types: regular dense; centered dense; centered star; segmented; pearl collar; dispersed network (Fig. 1).

A "regular dense" network is highly and quite regularly connected, or at least displays a dense core with some few isolated points. Its density is high but the index of betweenness centralization is low.

A "centered dense n̈etwork has the same general shape as the previous one but it has a high betweenness centralization index, which reveals the presence of a central alter connected to most of the others.

A 'centered star' is a network that is centered but has several small parts, and a lower density than the previous type. It looks like a star with a heart and branches.

A "segmented" network refers to a network distributed within several relatively large and dense components and some isolated points.

A network classified as "pearl collar" is elongated, its diameter is relatively high, and the values on the other indicators are not remarkable. This means that it is made of relatively long chains of ties.

A "dispersed" network appears very fragmented into small groups and isolated individuals.

These six types are exemplified in Fig. 1:

A regular dense network can be for example that of a young person living at parents' home and whose friends know his/her family or whose various groups of friends are intertwined. A centered dense network is typically that of a couple in which ego and his/her spouse share their family cliques and groups of friends, who are have in addition some links between them. In the case of a centered star, in contrast, these persons do not meet, and the spouse is the only connection between them (in addition to ego). A segmented network really leave these groups isolated the ones from the others, ego being their only common point (family is not connected to friends, and leisure, work and vacation groups, for example, are never mixed). In a pearl collar network the links between the different groups (family, leisure friends, workmates...) are provided by different people each time. This case shows that some different alters are making different bridges between different parts of the network which would be otherwise segmented. These particular alters are, typically, oldest friends from primary school or the same village who are connected with only some new friends, ex-partners who are still connected to old friends but not to new ones, a brother or sister who is connected to friends of the same age or co-workers when working in the same field, a best friend who is connected to one of the vacation friends, or a partner who for some reason cannot be presented to family but meets one friend. Sometimes they were introduced by ego, but sometimes other ties around them vanished,

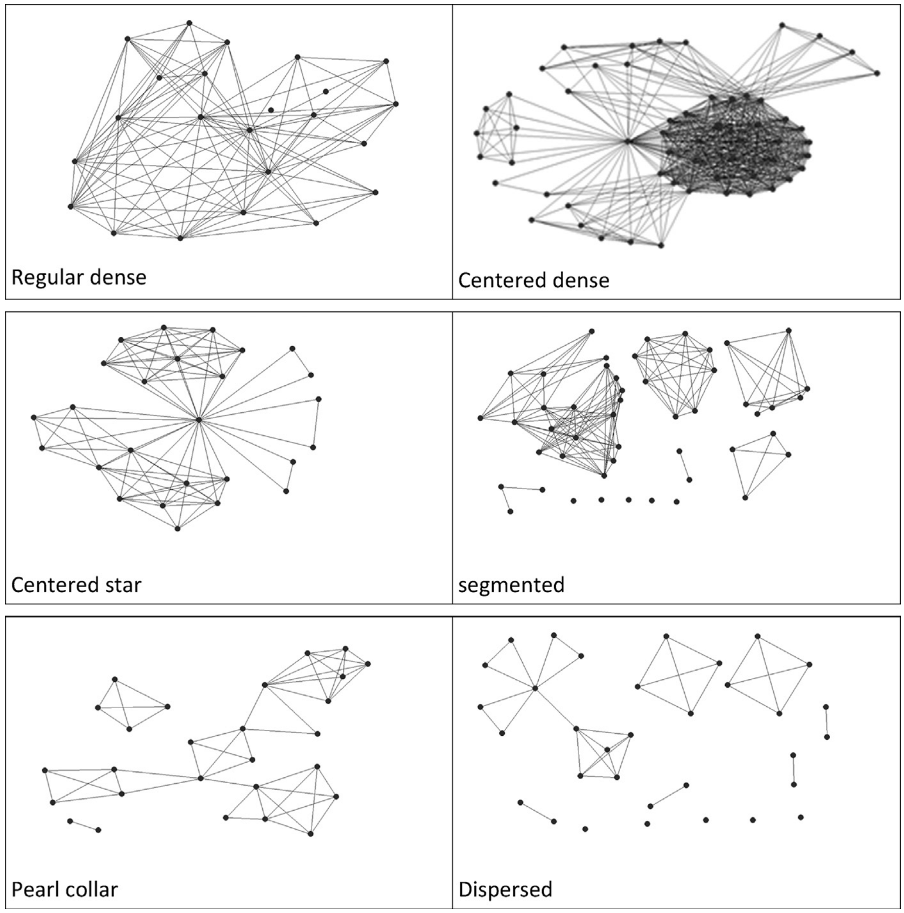

Fig. 1. Network graphs from our dataset, illustrating the 6 types.
Caption: nodes represent Alters, edges represent ties between Alters (Ego is not represented on his/her personal network graph).

for example when leaving far away. They are typically situated on a plurality of "structural holes" in the network (Burt, 1992). This also shows that what Granovetter calls the "forbidden triad"[2]; may occur in some situations.

Finally, in a dispersed network we have isolated units or groups, very few ties are built between the alters and ego really dissociates his/her relationships, one by one or by couples.

4.2. A "translation" into indicators

Having reached this stage, we had two objectives:

- Define a limited number of indicators ensuring the best way to arrive at this empirical typology.
- Try to reconstruct the typology in the form of a decision tree, which would automate the classification and reach a broader scope of the method.

To address the first question, we tried several statistical tools and finally used a discriminant analysis because it is the technique that best separates categories. We used it with six descriptive variables: density, betweenness centralization, modularity, the number of components (including isolated nodes), the proportion of alters in the largest component, and finally the diameter. To conduct this analysis, we selected the networks we considered most reliable, that is to say the networks easily attributable to each of the six particular kinds. The program then calculates the discriminant functions that can be used to classify additional networks. Secondly, we asked the program to assign a class (type) to other networks. The program offers each of these networks probabilities of assignment in each type. Our responsibility is ultimately to make the assignment decision for borderline cases.

This analysis allowed us to find the allocation of 214 networks out of 287, or 74.6%.

To answer the second question, we brought in a series of cuts on the variables in a given order. The best result was obtained with only four variables. The following decision tree gives the type of reconstruction algorithm (Fig. 2).

This segmentation begins by separating the cases where there is a very central alter from those where centrality is more homogeneous. This central alter is often (but not always) a spouse. Sometimes it is a sister or a friend who plays this role by sharing most links with ego. But sometimes a spouse is not at all in such a central position, when ego reserves for himself some of his solo relationships.[3] Then the modularity indicator separates cases of networks with few dense parts from cases where dense parts are more numerous. The denser parts are often related to the existence of social groups (families, cohesive groups of friends, etc.). The modularity indicator captures a portion of the information that is

[2] If person A has a strong tie to both B and C, then it is unlikely for B and C not to share a tie. See Granovetter (1973).

[3] Here we find the contrast Bott noticed, between couples in which each partner keeps his/her respective relationships, and couples who leave their former environments and build a new network together (Bott, 1957).

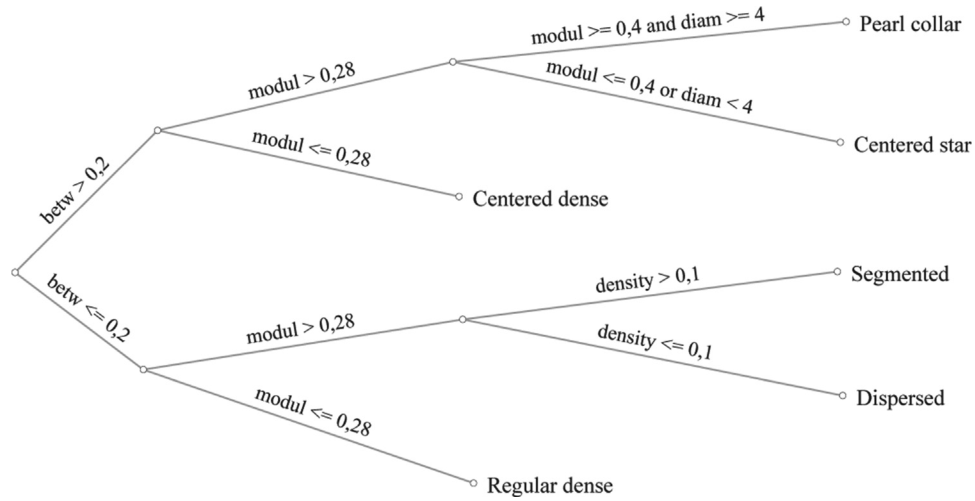

Fig. 2. Decision tree reproducing the best allocation in the 6 types Caption: When betweeness centrality is superior to 0.2, modularity is superior to 0.28, modularity is also superior to 0.4 and diameter is superior to 4, then the type is “Pearl collar”.

usually obtained by the calculation of density, without being too dependent on the size. That’s why we make it play an important role in this typology.

The combination of modularity (at a higher threshold) and diameter can distinguish the “pearl collar” type, which includes several distant central alters, from the “centered star” type, which includes only one. For slightly centered networks, the use of density can separate dispersed networks from those comprising several dense modules (“segmented” type).

This segmentation by the combination of thresholds reflects 205 cases out of the 287, or 71.4% of the 6 types.

These two steps allow to confirm the relevance of these six types.

4.3. *Assessing the typology*

One way to see if the criteria specify the types well is to look at the distribution of variable values in different types. The line in the center of each box represents the median and the limits of the box, respectively the first and third quartile of the distribution (Fig. 3).

The betweenness centralization allows to discriminate accurately the two centered types (centered dense and centered star) with respect to others; the modularity allows to isolate the two dense types (centered and regular); the density discriminates certain types like segmented and dispersed; and the diameter highlights the specificity of pearl collars (which could otherwise overlap with the “centered star” networks). This is the combination of these indicators which shows its relevancy.

The tension between sociological complexity closest to reality on the one hand, and methodological effectiveness on the other hand, continues to bother researchers. How widely this typology can be applied, and how few indicators are necessary for reproducibility to other data, will depend on the researchers’ goals. Indeed, the variables of betweenness centralization, modularity, diameter and density can be enough to guarantee a good ranking.

An analyst who would use our typology could, based on a fairly simple combination of these indicators, possibly by moving the thresholds, find the six types that are both well identified by the method and convincing in social science perspective. This relevance can be further tested by measuring correlations with well-known variables, namely the social classes.

5. Are the structural network types correlated with individual attributes of egos?

We can now test the correlations between these types and some sociological variables. Level of education is known as highly discriminating for a young population like that of our investigation. Here we have three different school courses: a general degree in economics and social sciences, leading to long studies; a vocational baccalaureate, leading to a rather rapid professional integration or short higher education; and finally a population of trainees in vocational integration, so a lower grade level than the “baccalauréat”. In the French school system, these courses are correlated with social origin and the future employability of students, from the more privileged for the first sector to the less for the third.[4] We tested other indicators (including gender and survey wave) but they didn’t prove significant.

5.1. *Types of education*

There are proportionally more “pearl collar”, “segmented” and “dispersed” networks in the most educated sector. There are more “centered star” and “regular dense” ones in the least educated sector. Respondents in the intermediate sector are distinguished by a high level of c̈entered dense” networks and a low level in the “pearl collar” ones. There is therefore a greater segmentation and dispersion of the networks of the most qualified young people, and greater density and centralization of those of less skilled youths (Table 1).

We can refine this result by measuring the correlation with social class of origin, measured by ego’s parents’ occupation.

The results point in the same direction: respondents from the upper class are more often classified in “pearl collar” or “dispersed” types, and more rarely in “centered dense” one, while the working classes are more commonly found in r̈egular densëtype.

Again, the social hierarchy corresponds to a contrast between the dense networks of the working classes and more fragmented networks of the upper classes. In sociological terms, this means more homogeneous and redundant social resources for the poor,

[4] The occupational category generally completes these factors; but since this survey involves young people who in many cases are still studying, this variable is less convenient.

and for the educated a broader, more segmented social environment.

Cross-tabulation by gender shows no interesting difference between the types, confirming other findings regarding gender and the characteristics of networks.

One can also study the link between residential status and types of networks, which allows the analyst to differentiate between respondents who live alone and those who live with at least one other person.

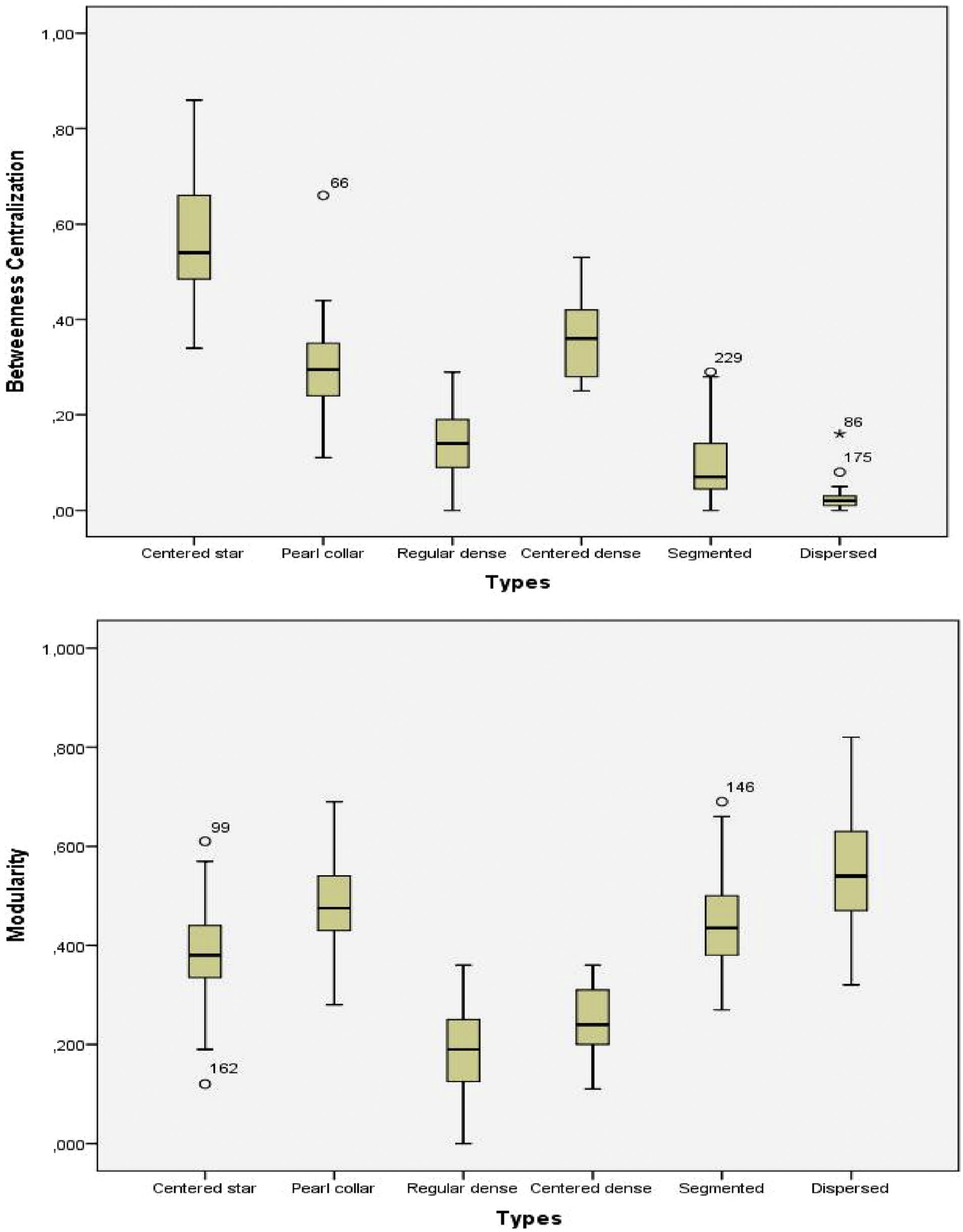

Fig. 3. Boxplots of the distribution of variable values for the 6 types.
Caption: The line in the center of each box represents the median and the limits of the box, respectively the first and third quartile of the distribution.

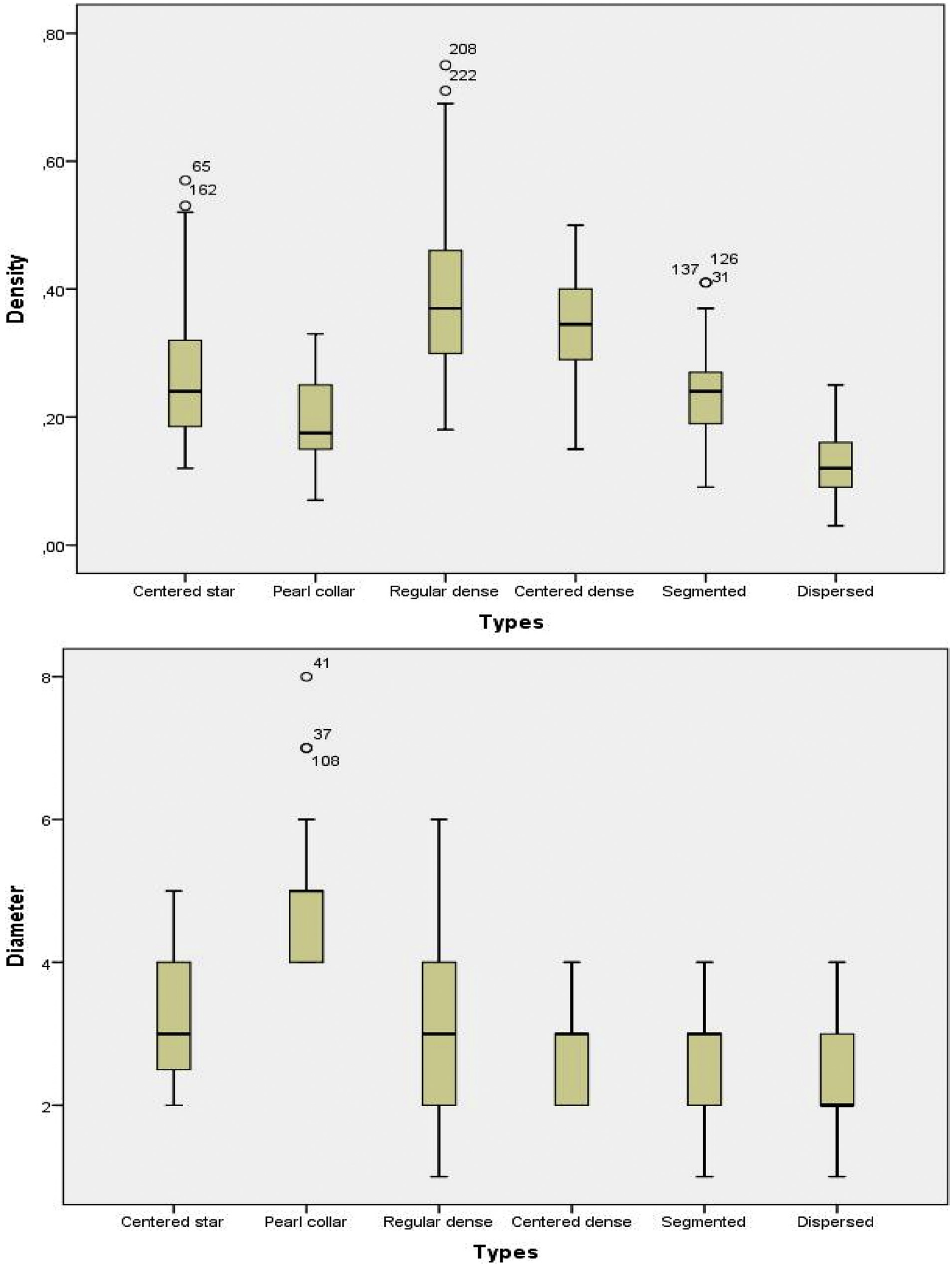

Fig. 3. (*Continued*)

Table 1
Distribution of the cases of our dataset in the 6 types by educational course.

	Centered star	Pearl collar	Regular dense	Centered dense	Segmented	Dispersed	Total
General degree in economics and social sciences	27	**21**	21	11	**27**	**18**	125
	21.6%	**16.8%**	16.8%	8.8%	**21.6%**	**14.4%**	100.0%
Vocational baccalaureate	21	7	25	**21**	18	4	96
	21.9%	7.3%	26.0%	**21.9%**	18.8%	4.2%	100.0%
Trainees in vocational integration	**19**	6	**25**	6	7	3	66
	28.8%	9.1%	**37.9%**	9.1%	10.6%	4.5%	100.0%
Total	67	34	71	38	52	25	287
	23.3%	11.8%	24.7%	13.2%	18.1%	8.7%	100.0%

$\chi^2 = 32.9$, $p < 0.01$.

Table 2
Distribution of the cases of our dataset in the 6 types by residential situation.

	Centered star	Pearl collar	Regular dense	Centered dense	Segmented	Dispersed	Total
Living in parents' home	20	12	37	16	31	6	122
	16.4%	9.8%	30.3%	13.1%	25.4%	4.9%	100.0%
Living alone	3	9	12	4	9	14	51
	5.9%	17.6%	23.5%	7.8%	17.6%	27.5%	100.0%
Living as a couple	44	13	22	18	12	5	114
	38.6%	11.4%	19.3%	15.8%	10.5%	4.4%	100.0%
	67	34	71	38	52	25	287
Total	23.3%	11.8%	24.7%	13.2%	18.1%	8.7%	100.0%

$\chi^2 = 59.3$, $p < 0.01$.

5.2. Residential status

Those who live alone tend to have dispersed networks. Those living with a partner as a couple tend to have networks of the 'centered star' type, confirming that the couple's life mostly gathers the network together, especially around the partner.[5] This particular position of the spouse is also at the heart of the demonstration by Backstrom and Kleinberg (2014) (Table 2).

Thus, the most centered networks in our categories ("centered star" and "centered dense") are very clearly associated with couples. Dense networks are associated with working-class origins and low levels of education. More dispersed networks are associated with intermediate or higher origins, the highest levels of education, and solo living.

6. Conclusion

A typology is a way to compare networks in a systematic way. As we saw from the other experiments presented at the beginning of this article, some authors choose to build their typology taking into account both the structure and characteristics of people or relationships. In this article we have tried using only the structural characteristics of the network to start with, thus following an inspiration close to structural interactionism (Degenne and Forsé, 1999; White, 2008). We sought to build an economical, synthetic method, with little dependence on "black boxes" or algorithms. The experiment is positive to the extent that we are able to define a parsimonious algorithm. The result is a typology of six categories that classify almost three quarters of our survey networks.

The personal networks our study has yielded are a complex and heterogeneous corpus, but four structural features can account for much of this complexity (betweenness centrality, modularity, diameter and density). Together, they indicate whether or not the network has a central individual who is connected with almost all the alters cited by the respondent; and whether the network has a high connectivity, or is separated into several components or communities, marking the fact that the respondent manages access to separate universes.

The first point refers to the person's lifestyle, whether he or she shares their life with another person, usually a spouse. Living alone or with a partner strongly influences the network structure. When the partner is connected with all the alters, you can almost say that it is the couple (the duo), and not the two individuals, that is the social networking entity.

The second point is at the basis of social capital theory. This theory envisages the personal network as a resource that is all the more effective as the network provides access to various social circles in which ego may find partners willing to provide assistance in different circumstances (Burt, 1992, 1995; Lin, 1995, 2001; Granovetter, 1973, 1974).[6] The level of education and social origin also significantly influence the structure of personal networks. We confirm this major trend, though structural measures are in principle disconnected from social contexts. For sure, we can notice some exceptions: some biographical experiences may counterbalance the weight of a lower social background and contribute to increase the network diversity and dispersion; also, some couples may keep their friends separate. The complexity of the phenomenon must be specified in more detailed analyses of biographical factors. But our structural analysis proves its relevance by testing these well-known variables. Here, we wanted to propose a method for identifying personal network structures that allows us to go beyond the specific data and may possibly prove likely to be more widely applied.

In fact, our population is specifically composed of young French people who were aged 17–24 years at the first survey wave and 26–33 years at the fourth wave. But it is precisely in this period of life that young people leave school and their family to find a job,

[5] We could also gather ego and his/her spouse in an analysis in terms of household. Yet all couples do not share their networks and it seems more interesting to look closely at these distinctions.

[6] Work on networks and social capital is organized around three principal authors: – Burt favors network resources in competitive situations. He focuses on the network structure, structural holes that are a sign of access to various universes (Burt, 1992). – Lin emphasizes the alters. He favors the wealth of resources available to alters and their tendency to enable ego to enjoy them (Lin, 2001). – Granovetter emphasizes the nature of links between ego and alters: weak links make it possible to access fresh information while strong links provide support that may be more available but holds less promise of original resources (Granovetter, 1973, 1974).

form a couple, have their first child for some of them, and generally move from a teenage sociability to an adult one, more intensive.

Precisely because these networks are constantly changing, it was a challenge to find generic features of these networks that were sufficiently abstracted from the contexts. The results seem encouraging enough to provide a method for other types of data. It is an experience that we are now pursuing, especially using online networks. The main interest of a purely structural typology is not only on itself, but is also to provide guidance on social situations without other information than the network itself. This can be very useful for the analysis of data from digital traces for example, in which very little social information is gathered. The typology has the advantage of being based on a small number of parameters, thus it is more "economical", less dependent on contexts and kinds of data, and more easily reproducible. The use of modularity in particular was very relevant to better manage variations in size. However, it is a first step towards the detection of social situations from network characteristics. One of our objectives is thus to see whether our approach can be reproducible and whether our typology can characterize personal networks in general.[7] Network size will be a central issue in the implementation of our typology to data that are not restricted to small or medium sized networks that are typically built in usual personal networks studies and include a great part of strong ties. Only replication on other surveys will provide answers. This gives us hope that further work could confirm the generality of these results and thus provide arguments to the view that a small set of typical forms would suffice to summarize the complexity of personal network structures. We hope they will allow all users to better understand the social situations through the information provided from the network structures.

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[7] We started to test the typology on a corpus of thousands of Facebook networks with colleagues specialists of such data. The first results seem promising.